

Earth-Observation Data Processing: First report - L1B Test

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The task involved applying Level-1B radiometric corrections (equalization and restoration) to remove sensor noise and convert the raw digital data into accurate physical values.

The results obtained were validated by comparing them with the outputs provided by the teacher.

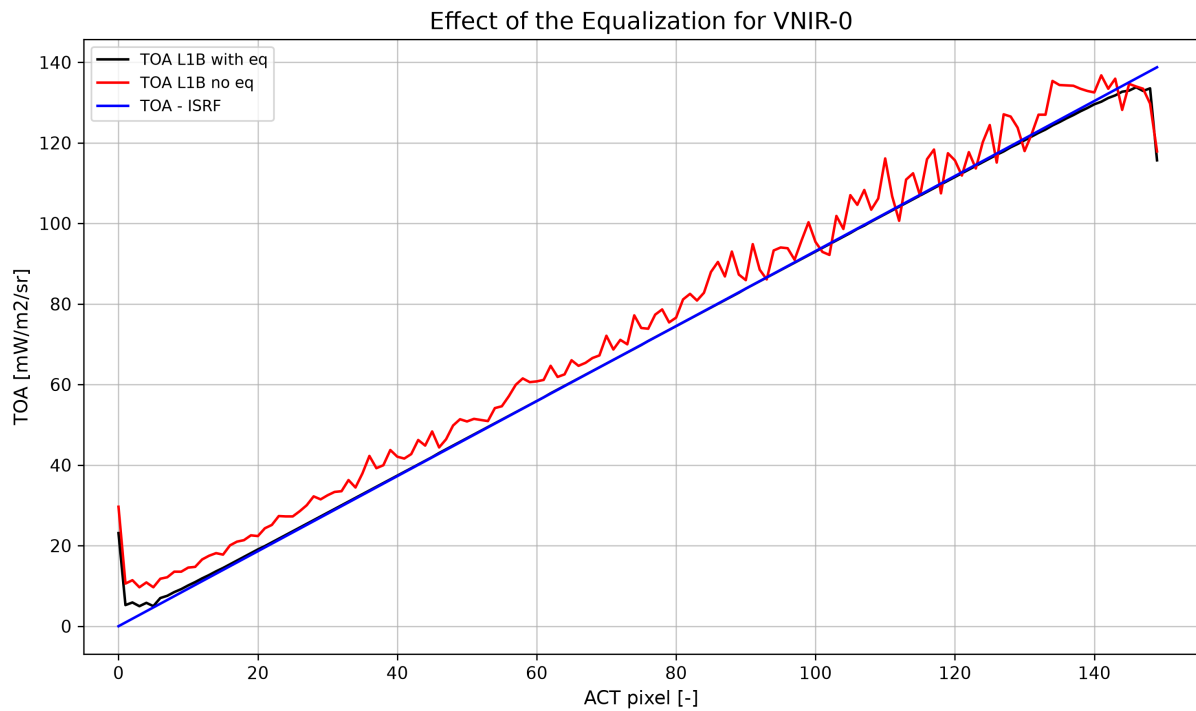


Figure 1: Comparison between equalized and non-equalized data, together with the ideal case.

Firstly, both the non-equalized and the equalized case went through restoration to convert the digital numbers into radiance. Additionally, it can be seen that the unequalized data (red) presents significant noise, caused by two particular effects that differ from pixel to pixel: the Dark Signal and the PRNU. After the equalization, the corrected signal (blue) is much smoother and aligns closely with the ideal radiance (black).